

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

END SEM EXAMINATION - Summer Supplementary 2022

Course: B. Tech in Electronics and Telecommunication Engineering

Sem: IV

Subject Name: Analog Communication Engineering

Subject Code: BTEXC402

Max Marks: 60

Date:-

Duration:- 1 Hr.

Instructions to the Students:

1. **Draw suitable diagram wherever necessary.**
2. **Answer any Five questions**

- Q.1** **A** Explain in detail: Block schematic of communication system. Explain different blocks of communication system in detail. 8
- B** Define: Simplex communication, Duplex communication, Broadcast communication and Point to point communication. 4
- Q.2** Define amplitude modulation. Derive expressions for sideband frequency, modulation index and instantaneous voltage equation for DSBFC wave. Also draw time domain and frequency domain (frequency domain) representation for the same. Also find out power relation between DSBFC wave and carrier wave. 12
- Q.3.** What is modulation? Derive the expression for Modulation index for FM 12
- Q.4** **A** Two resistors 20 k Ω and 50 k Ω are at room temperature (290 K). Calculate for bandwidth of 100 KHz, the thermal noise for the following conditions: 8
- i) For resistor 20 k Ω
- ii) For resistor 50 k Ω
- iii) For two resistors in series
- iv) For two resistors in parallel.
- B** An amplifier has a bandwidth of 4 MHz with 10 k Ω as the input resistor. Calculate the rms noise voltage at the input to this amplifier if the room temperature is 25°C. 4
- Q.5** **A** When the modulating frequency in an FM system is 400 Hz and the modulating voltage is 2.4 V, the modulation index is 60. Calculate the maximum deviation. What is the modulating index when the modulating frequency is reduced to 250 Hz and the modulating voltage is simultaneously raised to 3.2 V? 6
- B** Write a short note on i) practical diode detector ii) phase discriminator 6
- Q.6** **A** An A.F. signal $20 \sin(2\pi \times 500 t)$ is used to amplitude modulate a carrier of $50 \sin(2\pi \times 10^5 t)$. Calculate : 10
- (i) Modulation index
- (ii) Sideband frequencies
- (iii) Amplitude of each sideband frequency
- (iv) Bandwidth required
- (v) Total power delivered into a load of 600 Ω .
- B** Explain the Carsons rule 2

***** End *****